**Reading Notes for Week 8(Selected)**

## Chapter 11 Sampling Algorithms

### Goal of sampling methods:

- Simulate some distribution.
- Find the expectation w.r.t. A probability distribution
- Sample from a proposed distribution and estimate some statistical quantities of the target distribution.

### Inverse Transform Method

- Assumptions
  - Must know how to compute the CDF of the target distribution,
  - The CDF must be bijection(invertible).
- Procedure
  - Sample from z ~ U(0,1)
  - Compute F*(z) as a sample from target distribution.
- Proof Sketch
  - If CDF of X is F(.), then P(F(X)<=x) = x, i,e F(X) is U(0,1)
- Can be generalized to multiple-dimension using conditional distribution or maybe graphical model assumptions. (Three-triple models).
  - Want to sample multivariate gaussian x,y
    - First sample u ~ Y using inverse transform method
    - Sample z ~ X|y at y using inverse transform method
    - (u,z) is what we want.

### Rejection Sampling Method

- Motivation
  - The rejection sampling framework allows us to sample from relatively complex distributions, subject to certain constraints.
  - We don't need explicit form of CDF of target distribution.
- Procedure
  - Find a well-chosen proposed distribution q(x) and k such that k q(x) >= p(x) / M pointwise.
  - Sample u ~ U(0,1)
  - Sample x ~ q
  - Apply rejection rules to decide whether to keep x or not.

### Adaptive Sampling Method

- Motivation
  - Rejection Sampling often has high rejection rate due to poor choice of proposed distribution function.
  - For log-concave probability density,computing its envelopes(upper bound, just like kq(x)) is easy.
- Procedure

- Establish a set of fixed points and evaluate the log-density, h(x)=log(p(x)), and derivative of the log-density on the fixed points.
- Use these function evaluations to construct a piecewise-linear, upper bound for the log-density function, h+(x) via supporting tangent lines of the log-density at the fixed points.
- Let g+(x)=exp(h+(x)). Since h+(x) is piecewise linear, g+(x) is piecewise exponential. Because of this property, sampling y ~ g+(x) is straightforward.
- Pick u ~ U(0,1)
    - If u <= exp(h(y)-h+(y)) , accept y.
    - Else, draw another sample from g+.
- For any y rejected by the above criteria, y may be added to the initial set of fixed points and the piecewise linear upper bound, h+, adaptively updated.

**Monte Carlo Methods**

- Markov Chain
    - A random process that has very nice properties
        - Markov Property
            - Futures are conditionally independent of the current given the past.
        - Probability of a Path
            - Just use the chain rule of probability and markov properties
        - Irreducibility
            - Communicating class: The same concept as strongly connected components in the transition diagram.
            - A markov chain is irreducible if there is only one communicating class.
        - Recurrence
            - A state is recurrent if starting from this state, after some time, the process will go back to this state.
            - A communicating class is recurrent class if there is at least one recurrent state in this class.
        - Periodicity
            - A recurrent class is aperiodic if the period of at least one state is 1.
            - Or the state diagram has self-transition.
- MCMC
    - Motivations
        - Want to find the mode of the stationary distribution of a markov chain.
        - The actual stationary distribution is computationally impossible.
        - So we want to simulate a random variable that has the stationary distribution that we want to compute the mode from.
    - Procedure
        - Starts with any symmetric, irreducible transition matrix Q(proposed transition distribution).

    - Use some randomization to create a markov chain that is irreducible and aperiodic and thus has a stationary distribution.
    - Determine the transition probability of new markov chain to be created, which is parametrized by transition matrix P.
        - Suppose the chain is at i at time n(i.e. Suppose Xn=i. Pick a state j according to the proposal probability Q(i,j). This j is the candidate state to which your chain might move.
        - Define the acceptance ratio r(i,j)=pi(j)/pi(i)
            - If r(i,j) >=1, Set X_{n+1}=j
            - If r(i,j) < 1. Toss a coin that lands heads with change r(i,j)
                - If the coin lands heads, set X_{n+1} = j.
                - If the coin lands tails, set X_{n+1}=i.
        - Repeat all the steps, with X_{n+1} as the starting value.
    - The new chain is constructed by Q and is denoted by P. The new chain is irreducible because the proposal chain is irreducible. It is aperiodic because it can stay in place. So it has a steady state distribution.
    - Finally the new chain(sample chain) that we created will have the same stationary distribution as the proposed stationary distribution. Then we can find the mode of the stationary distribution based on this sample.
- Applications
    - Code Breaking, evaluate the performance of the decoder.